1. An electron experiences a force $(1 \cdot 6 \times 10^{-16}\ \text{N})\ \hat{i}$ in an electric field $\vec{E}$. The electric field $\vec{E}$ is :

(a) $(1 \cdot 0 \times 10^{3}\ \frac{\text{N}}{\text{C}})\ \hat{i}$

(b) $-(1 \cdot 0 \times 10^{3}\ \frac{\text{N}}{\text{C}})\ \hat{i}$

(c) $(1 \cdot 0 \times 10^{-3}\ \frac{\text{N}}{\text{C}})\ \hat{i}$

(d) $-(1 \cdot 0 \times 10^{-3}\ \frac{\text{N}}{\text{C}})\ \hat{i}$

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(c) $(1{\cdot}0 \times 10^{-3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{i}$

(d) $-(1{\cdot}0 \times 10^{-3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{i}$

(b) $-(1.0 \times 10^{3}\ \frac{N}{C})\hat{i}$

5. An isolated point charge particle produces an electric field $\vec{E}$ at a point 3 m away from it. The distance of the point at which the field is $\frac{\vec{E}}{4}$ will be **1**

(a) 2 m

(b) 3 m

(c) 4 m

(d) 6 m

5. An isolated point charge particle produces an electric field $\vec{E}$ at a point 3 m away from it. The distance of the point at which the field is $\frac{\vec{E}}{4}$ will be **1**

(a) 2 m (b) 3 m

(c) 4 m (d) 6 m

(d) 6 m

1. The magnitude of the electric field due to a point charge object at a distance of 4.0 m is 9 N/C. From the same charged object the electric field of magnitude, $16\,\frac{\text{N}}{\text{C}}$ will be at a distance of **1**

(a) 1 m (b) 2 m

(c) 3 m (d) 6 m

1. The magnitude of the electric field due to a point charge object at a distance of 4.0 m is 9 N/C. From the same charged object the electric field of magnitude, 16 $\frac{N}{C}$ will be at a distance of **1**

(a) 1 m (b) 2 m

(c) 3 m (d) 6 m

(c)
3 m

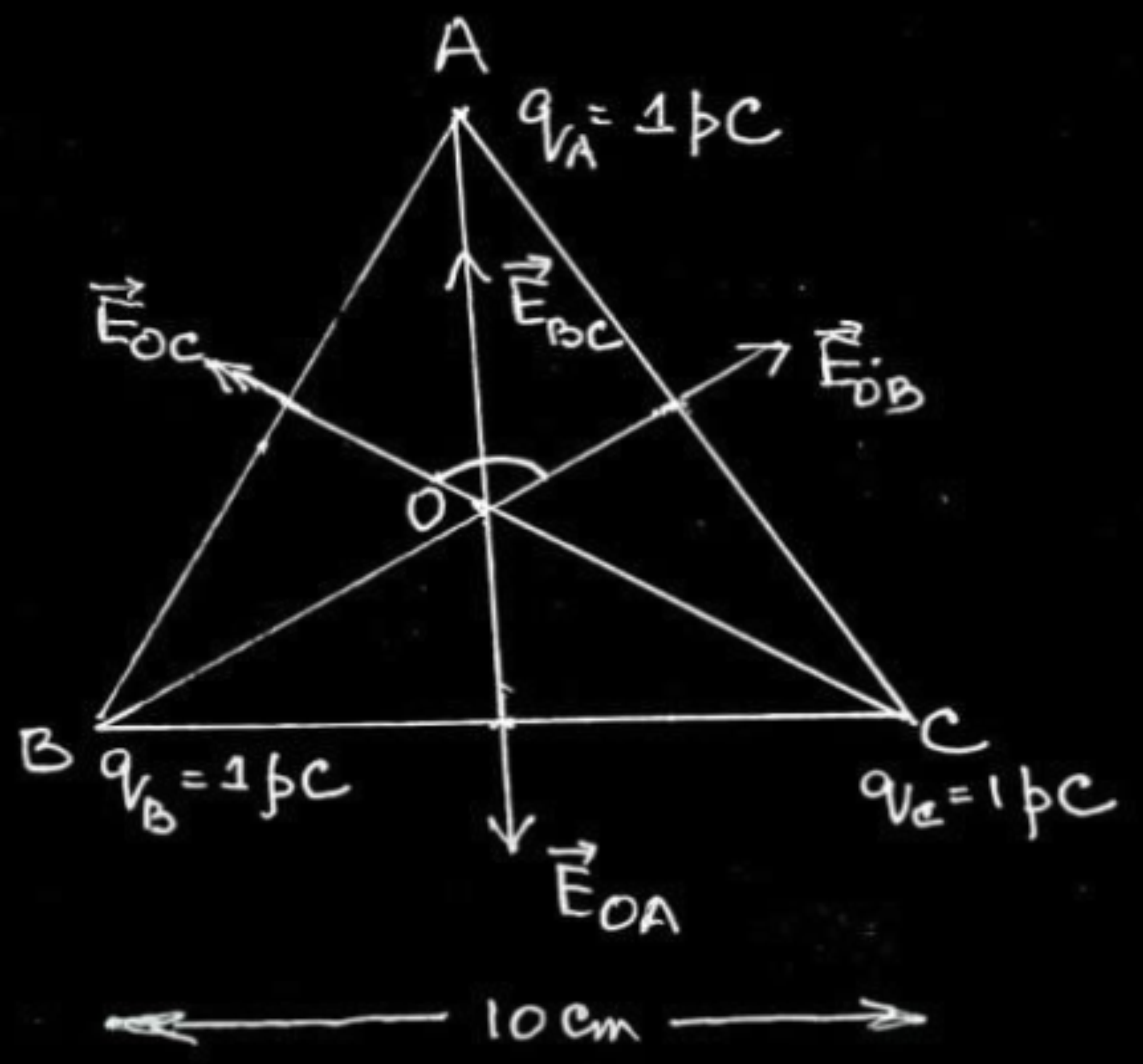
31. (a) (i) Why do we use a small test charge to measure an electric field ?

(ii) A small stationary positively charged particle is free to move in an electric field. In which direction will it begin to move ?

(i) So that it does not disturb the distribution of charges whose electric field is to be measured. Otherwise field will be different from the actual field.	1
(ii) It will move in the direction of electric field.	1

18. (b) Three point charges, 1 pC each, are kept at the vertices of an equilateral triangle of side 10 cm. Find the net electric field at the centroid of triangle.

Finding net electric field at centroid	2



1

$$q_A = q_B = q_C = 1pC$$

$$AO = BO = CO = r$$

$$\left|\vec{E}_{OA}\right| = \left|\vec{E}_{OB}\right| = \left|\vec{E}_{OC}\right|$$

$$\vec{E}_{BC} = \vec{E}_{OB} + \vec{E}_{OC}$$ ½

$$E_{BC} = \sqrt{E_{OB}^2 + E_{OC}^2 + 2E_{OB}E_{OC}\cos 120^0}$$

$$E_{BC} = E_{OB} \quad , \quad \vec{E}_{OA} = -\vec{E}_{BC}$$

Net electric field $\vec{E}_O = \vec{E}_{OA} + \vec{E}_{BC}$ ½

$$\vec{E}_O = 0$$

Alternatively

$$E_{OA} = E_{OB} = E_{OC} = 2.7\ NC^{-1}$$

$$E_{BC} = \sqrt{E_{OB}^2 + E_{OC}^2 + 2E_{OB}E_{OC}\cos 120^0}$$ ½

$$= E_{OB}$$

As $\vec{E}_{BC} = -\vec{E}_{OA}$

$$\vec{E}_{BC} + \vec{E}_{OA} = 0$$ ½

Net electric field is zero.

Alternatively

$$\left|\vec{E}_{OA}\right| = \left|\vec{E}_{OB}\right| = \left|\vec{E}_{OC}\right|$$

Electric field vectors are making an angle of 120^0 with each other. They make a closed polygon. So vector sum of all electric field vectors will be zero.

$\vec{E} = 0$ 2

1. Two charged particles P and Q, having the same charge but different masses m_P and m_Q, start from rest and travel equal distances in a uniform electric field $\vec{E}$ in time t_P and t_Q respectively. Neglecting the effect of gravity, the ratio $\left(\frac{t_P}{t_Q}\right)$ is :

(A) $\frac{m_P}{m_Q}$

(B) $\frac{m_Q}{m_P}$

(C) $\sqrt{\frac{m_P}{m_Q}}$

(D) $\sqrt{\frac{m_Q}{m_P}}$

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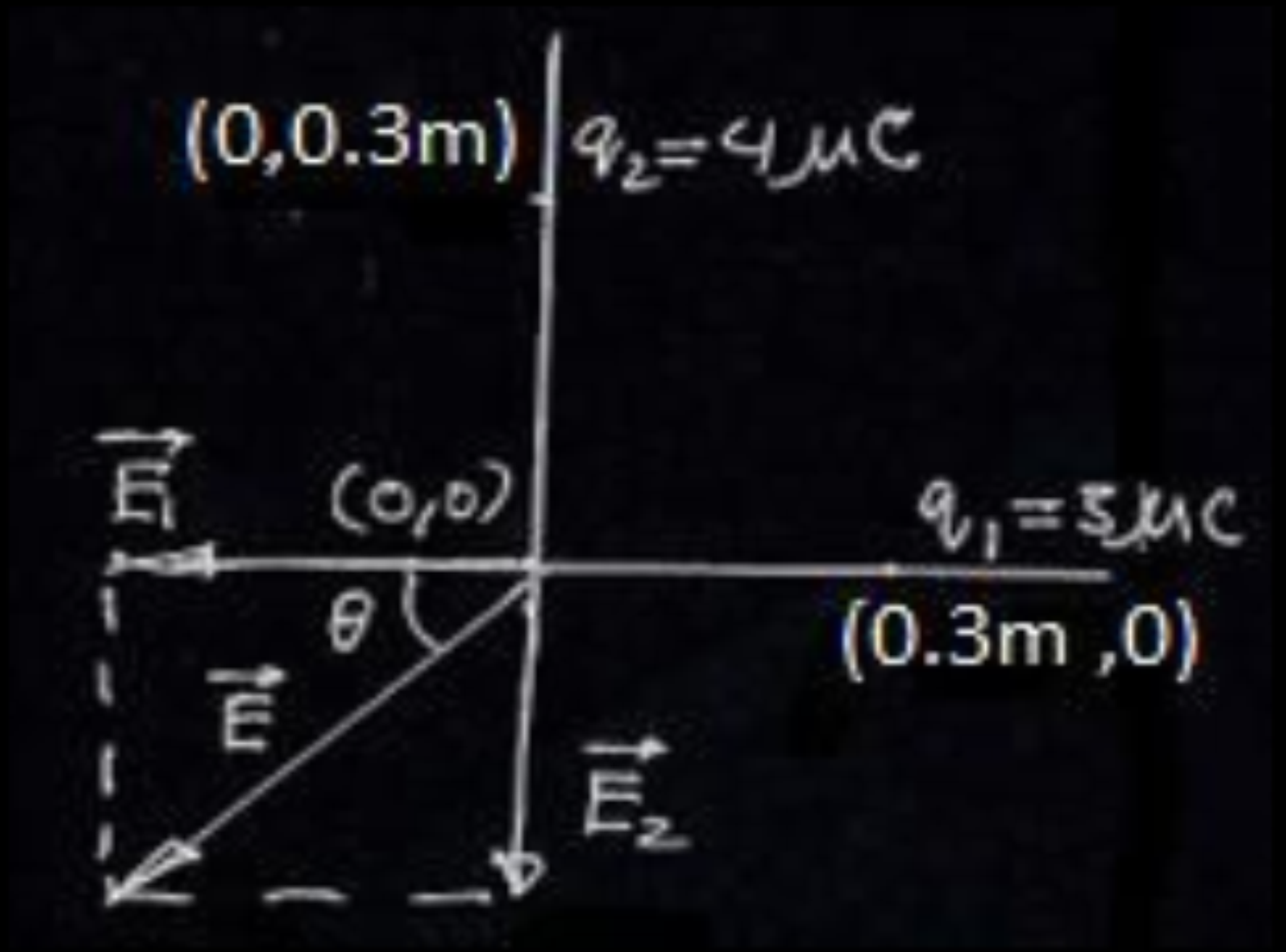
(D) $\sqrt{\frac{m_Q}{m_P}}$

(C) $\sqrt{\frac{m_p}{m_Q}}$

(b) Two point charges of 3 μC and 4 μC are kept in air at (0·3 m, 0) and (0, 0.3 m) in x-y plane. Find the magnitude and direction of the net electric field produced at the origin (0, 0). *2*

Finding the magnitude of electric field	1 ½
Finding the direction of electric field	½



$\vec{E}_1 = \frac{kq_1}{r_1^2}(-\hat{i}) = \frac{9\times10^9\times3\times10^{-6}}{(0.3)^2}(-\hat{i}) = 3\times10^5(-\hat{i})\,\mathrm{NC}^{-1}$ ½

$\vec{E}_2 = \frac{kq_2}{r_2^2}(-\hat{j}) = \frac{9\times10^9\times4\times10^{-6}}{(0.3)^2}(-\hat{j}) = 4\times10^5(-\hat{j})\,\mathrm{NC}^{-1}$ ½

$\vec{E} = \vec{E}_1 + \vec{E}_2$

$E = \sqrt{E_1^2 + E_2^2}$

$E = 5\times10^5\ \mathrm{NC}^{-1}$ ½

$\tan\theta = \frac{4}{3}$

$\theta = \tan^{-1}\left(\frac{4}{3}\right)$ inclination with respect to the x-axis (in III quadrant). ½

(a) (i) Define electric field. Write its SI units. Derive an expression for electric field due to a point charge at a point. Why must the test charge be as small as possible ?

(ii) An electron and an α-particle, both are kept in the same electric field $\vec{E} = E_1 \hat{i} + E_2 \hat{j}$. Find the force acting on them and their acceleration. *5*

(i)Defining electric field	1
SI units	½
Deriving expression for electric field	1
Reason for test charge to be small	½
(ii) Finding the force on electron & α-particle	½+½
Acceleration of electron & α-particle	½+½

(i) Electric field due to a change Q at a point in space may be defined as the force that a unit positive charge would experience if placed at that point. 1
S I unit is N/C

Alternatively- ½
SI unit is V/m

The force between a source charge Q and test charge q will be

$$\vec{F} = \frac{Qq}{4\pi\varepsilon_0 r^2}\hat{r}$$

The electric field due to the source charge

$$\vec{E} = \lim_{q\to 0}\left(\frac{\vec{F}}{q}\right)$$

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{r}$$ 1

- The test charge should be as small as possible so that the source charge remains at its original location. ½

(ii) Electric field $(\vec{E}) = E_1\hat{i} + E_2\hat{j}$ (Given)

Force experienced by the electron $\vec{F}_e = -e\vec{E} = -e\left[E_1\hat{i} + E_2\hat{j}\right]$ ½

Acceleration of the electron $\vec{a}_e = -\frac{\vec{F}_e}{m_e} = -\frac{e}{m_e}\left[E_1\hat{i} + E_2\hat{j}\right]$ ½

Force experienced by an alpha particle ½
$\vec{F}_\alpha = 2e\vec{E} = 2e\left[E_1\hat{i} + E_2\hat{j}\right]$

Acceleration of the alpha-particle ½
$\vec{a}_\alpha = \frac{2e}{m_\alpha}\left[E_1\hat{i} + E_2\hat{j}\right]$

3. A particle of mass m and charge q moving with velocity $v\hat{i}$ is subjected to a uniform electric field $E\hat{j}$. The particle will initially have a tendency to move in a circle of radius :

(A) $\left(\frac{mv}{qE}\right)$ in x-y plane

(B) $\left(\frac{mv^2}{qE^2}\right)$ in x-z plane

(C) $\left(\frac{mv^2}{qE}\right)$ in x-y plane

(D) $\left(\frac{mv}{qE^2}\right)$ in y-z plane

3. A particle of mass m and charge q moving with velocity $v\hat{i}$ is subjected to a uniform electric field $E\hat{j}$. The particle will initially have a tendency to move in a circle of radius :

(A) $\left(\dfrac{mv}{qE}\right)$ in x-y plane

(B) $\left(\dfrac{mv^2}{qE^2}\right)$ in x-z plane

(C) $\left(\dfrac{mv^2}{qE}\right)$ in x-y plane

(D) $\left(\dfrac{mv}{qE^2}\right)$ in y-z plane

(C) $\left[\dfrac{mv^2}{qE}\right] in\ X-Y\ plane$

(ii) Two point charges q_1 (= 16 μC) and q_2(= 1 μC) are placed at points $\vec{r}_1 = (3\text{ m})\hat{i}$ and $\vec{r}_2 = (4\text{ m})\hat{j}$. Find the net electric field $\vec{E}$ at point $\vec{r} = (3\text{ m})\hat{i} + (4\text{ m})\hat{j}$.

(ii)

$$\vec{E} = \frac{Kq}{r^2}\hat{r}$$

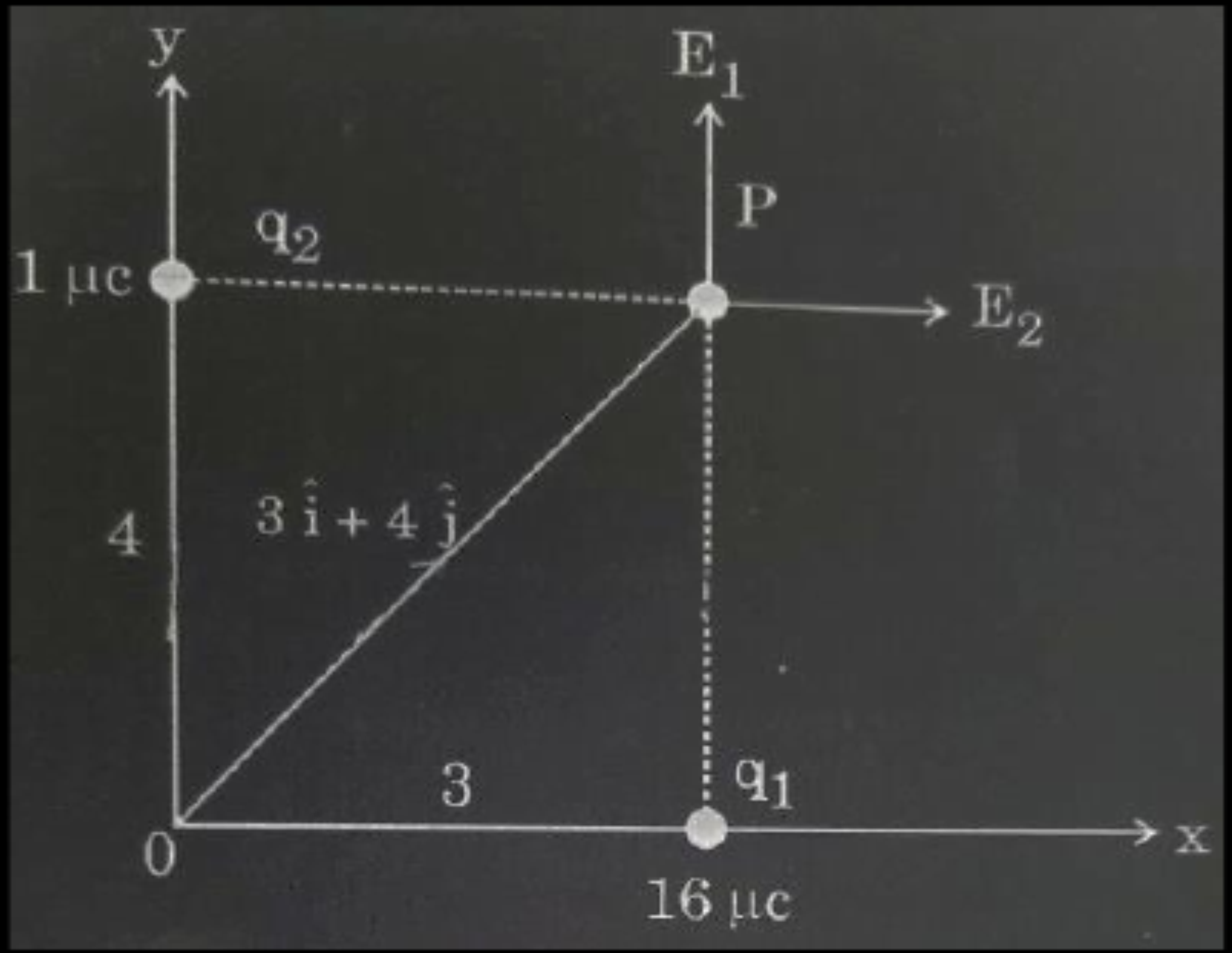
$$\vec{E}_1 = \frac{9\times10^9\times16\times10^{-6}}{(4)^2}\hat{j}$$ ½

$$= 9\times10^3\ \hat{j}$$ ½

$$\vec{E}_2 = \frac{9\times10^9\times1\times10^{-6}}{(3)^2}\hat{i}$$ ½

$$= 10^3\ \hat{i}$$

$$\vec{E}_{net} = (\hat{i} + 9\hat{j})\,10^3\ N/C$$ ½



NOTE: Award full credit of this part if a student finds magnitude and direction separately.

1. Two point charges Q and $-$ q are held ‘r’ distance apart in free space. A uniform electric field $\vec{E}$ is applied in the region perpendicular to the line joining the two charges. Which one of the following angles will the direction of the net force acting on charge $-$ q make with the line joining Q and $-$ q ?

(A) $\tan^{-1} \dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(B) $\cot^{-1} \dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(C) $\tan^{-1} \dfrac{QE}{4\pi\varepsilon_0 r^2}$

(D) $\cot^{-1} \dfrac{QE}{4\pi\varepsilon_0 r^2}$

1. Two point charges Q and $-q$ are held 'r' distance apart in free space. A uniform electric field $\vec{E}$ is applied in the region perpendicular to the line joining the two charges. Which one of the following angles will the direction of the net force acting on charge $-q$ make with the line joining Q and $-q$?

(A) $\tan^{-1}\dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(B) $\cot^{-1}\dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(C) $\tan^{-1}\dfrac{QE}{4\pi\varepsilon_0 r^2}$

(D) $\cot^{-1}\dfrac{QE}{4\pi\varepsilon_0 r^2}$

(A) $\tan^{-1}\dfrac{4\pi\varepsilon_o Er^2}{Q}$

1. Two charges –q each are placed at the vertices A and B of an equilateral triangle ABC. If M is the mid-point of AB, the net electric field at C will point along 1

(A) CA
(B) CB
(C) MC
(D) CM

1. Two charges –q each are placed at the vertices A and B of an equilateral triangle ABC. If M is the mid-point of AB, the net electric field at C will point along 1

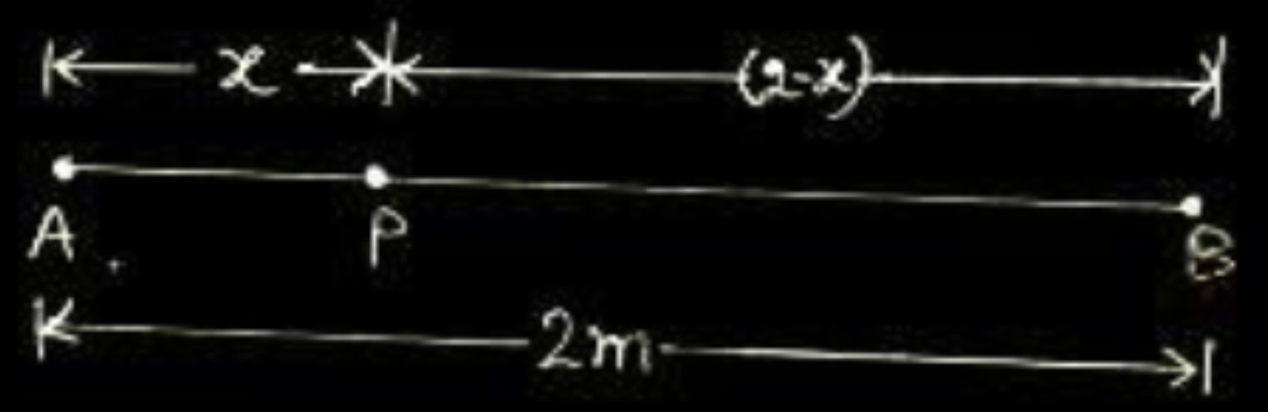
(A) CA (B) CB

(C) MC (D) CM

(D) CM

(b) Two point charges $q_1 = +1\ \mu C$ and $q_2 = +4\ \mu C$ are placed 2 m apart in air. At what distance from q_1 along the line joining the two charges, will the net electric field be zero ?

b)



$$E_{PA} = E_{PB} \quad ; \quad E = \frac{kq}{r^2}$$

$$\frac{kq_A}{x^2} = \frac{kq_B}{(2-x)^2}$$ ½

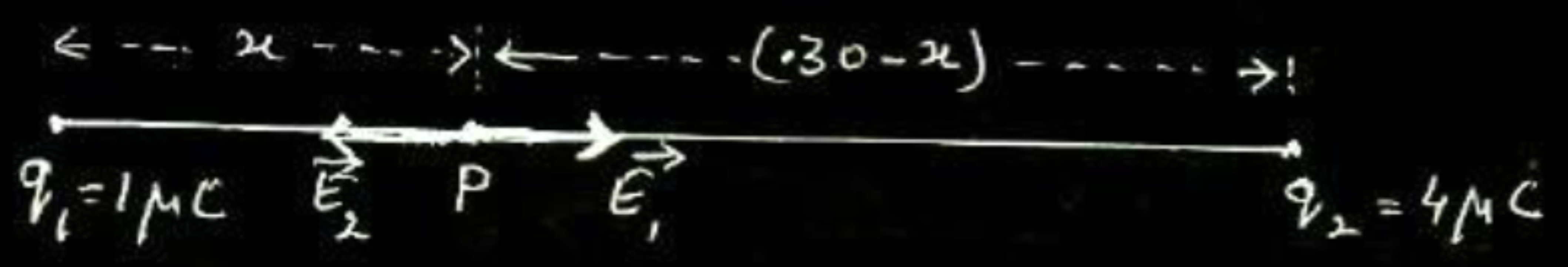
$$\frac{1}{x^2} = \frac{4}{2-x)^2}$$ ½

$$\frac{1}{x} = \frac{2}{2-x}$$ ½

$$x = \frac{2}{3}\ m$$ ½

(b) Two point charges of $+1\ \mu C$ and $+4\ \mu C$ are kept 30 cm apart. How far from the $+1\ \mu C$ charge on the line joining the two charges, will the net electric field be zero ?

(b)



½

$E_1 = E_2$

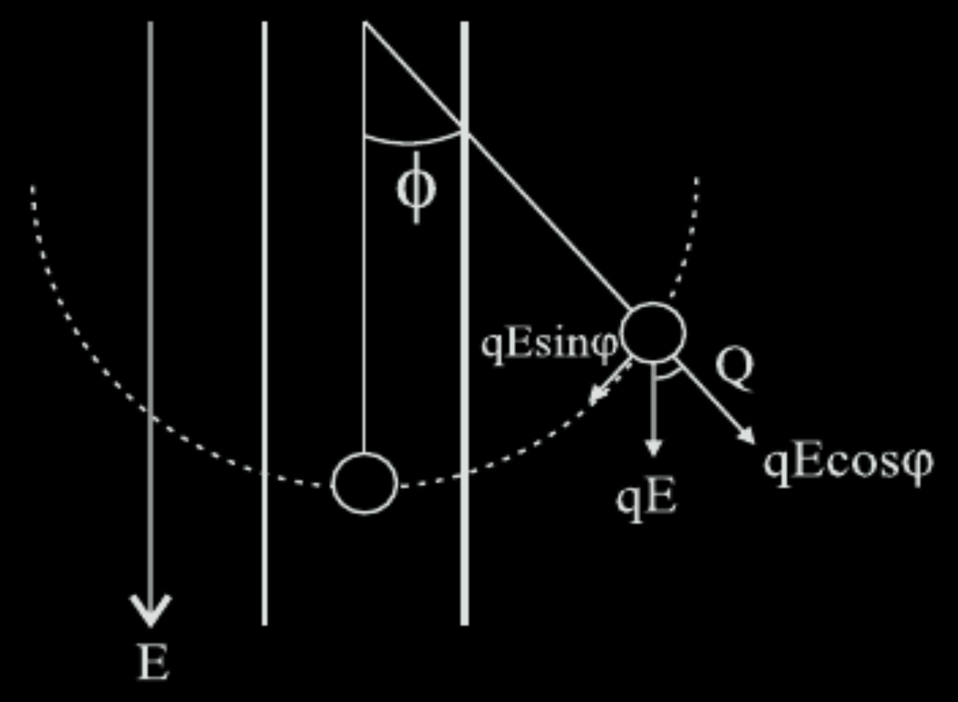
$$\frac{1}{4\pi\epsilon_o}\frac{1\times10^{-6}}{x^2} = \frac{1}{4\pi\epsilon_o}\frac{4\times10^{-6}}{(0.3-x)^2}$$ ½

$$(0.3-x)^2 = 4x^2$$ ½

$$0.3 - x = 2x$$ ½

$$x = 0.1m = 10\ cm\ (to\ the\ right\ of\ q_1)$$

A simple pendulum consists of a small sphere of mass m suspended by a thread of length l. The sphere carries a positive charge q. The pendulum is placed in a uniform electric field of strength E directed vertically downwards. Find the period of oscillation of the pendulum due to the electrostatic force acting on the sphere, neglecting the effect of the gravitational force.



½

Derivation

$F_r = -qE\sin\varphi$ (Restoring force) ½

$ma = -qE\sin\varphi$

when φ is small

$ma = -qE\varphi$

$$m\frac{d^2x}{dt^2} = -qE\frac{x}{l}$$

$$\frac{d^2x}{dt^2} = -q\frac{E}{m}\frac{x}{l}$$ ½

comparing with equation of linear SHM.

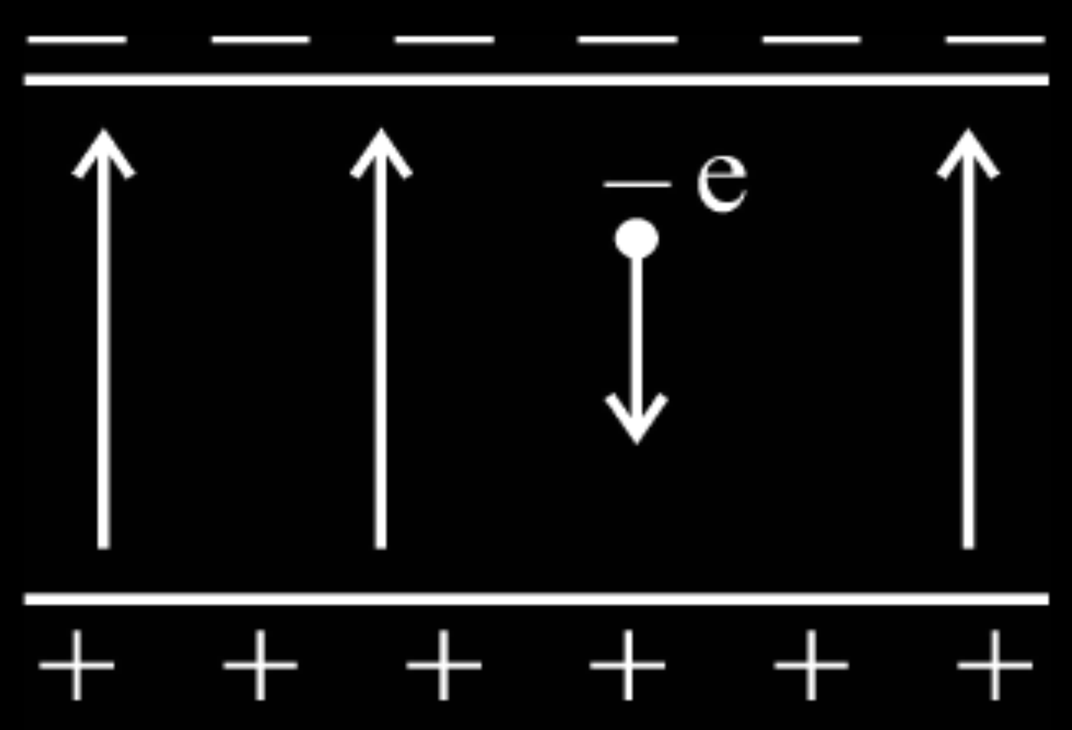
$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\omega = \sqrt{\frac{qE}{ml}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{ml}{qE}}$$ ½

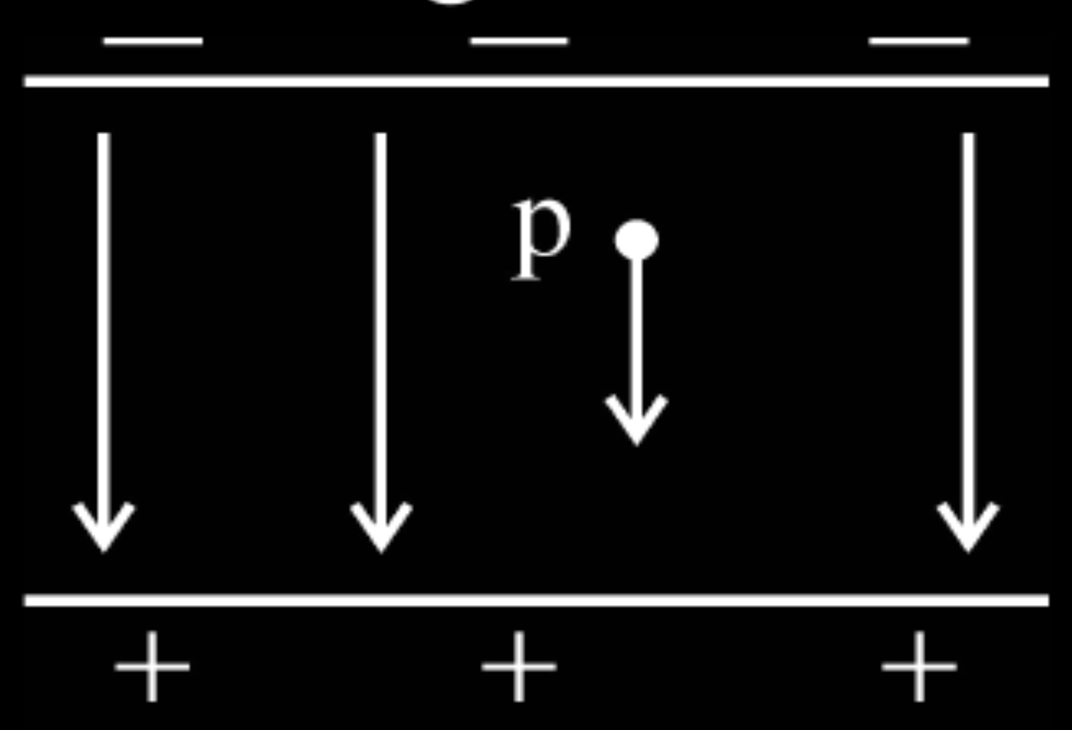
Alternatively - The student can use angular SHM expression also. Full marks to be awarded for correct answer even without intermediate steps.

An electron falls through a distance of 1.5 cm in a uniform electric field of magnitude 2.0×10^4 N/C (Fig. a)



(a)

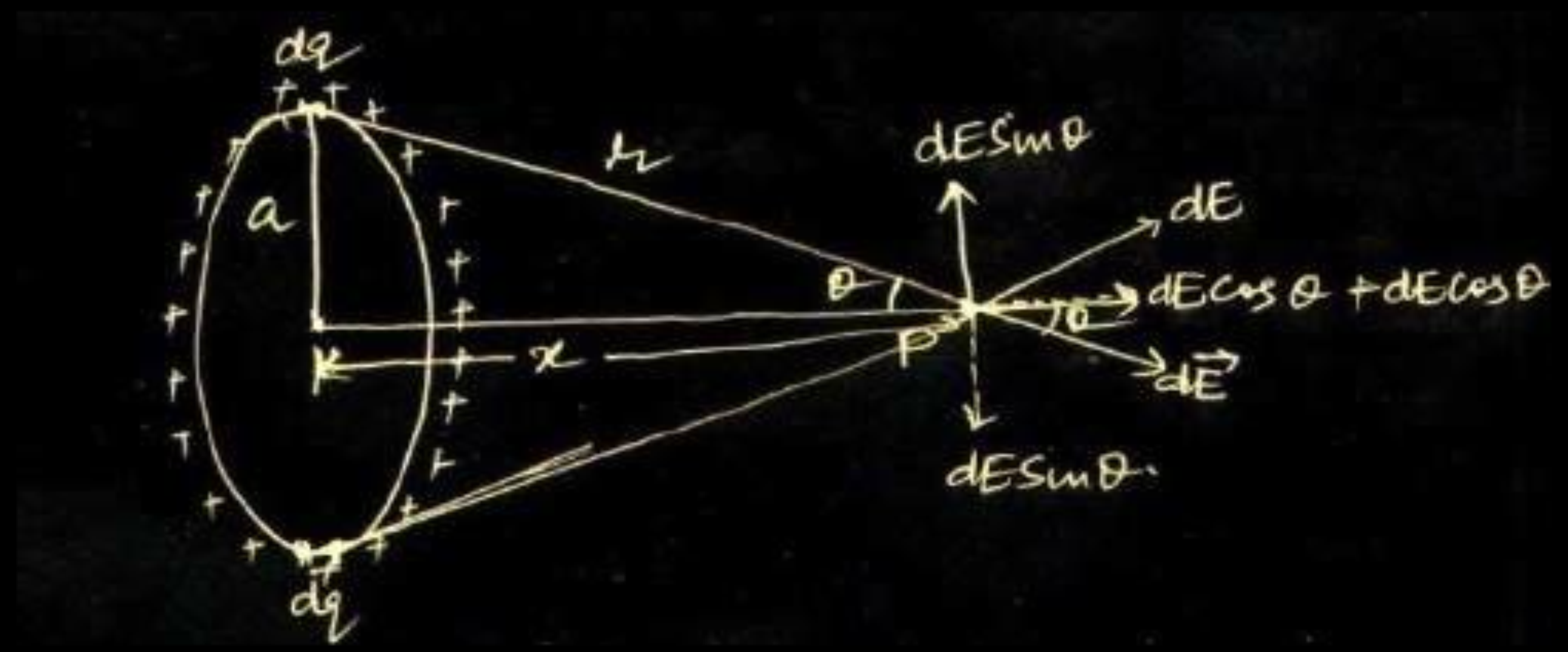
Calculate the time it takes to fall through this distance starting from rest.



(b)

If the direction of the field is reversed (fig. b) keeping its magnitude unchanged, calculate the time taken by a proton to fall through this distance starting from rest.

A charge is distributed uniformly over a ring of radius ‘a’. Obtain an expression for the electric intensity E at a point on the axis of the ring. Hence show that for points at large distances from the ring, it behaves like a point charge.



½

Net Electric Field at point P = $\int_o^{2\pi a} dE\cos\theta$

dE = Electric field due to a small element having charge dq ½

$= \frac{1}{4\pi\varepsilon_o}\frac{dq}{r^2}$

Let λ = Linear charge density

$= \frac{dq}{dl}$

$dq = \lambda dl$

Hence $E = \int_o^{2\pi a}\frac{1}{4\pi\varepsilon_o}\cdot\frac{\lambda dl}{r^2}\times\frac{x}{r}$, where $\cos\theta = \frac{x}{r}$ ½

$$= \frac{\lambda x}{4\pi\varepsilon_o r^3}(2\pi a)$$

$= \frac{1}{4\pi\varepsilon_o}\frac{Qx}{(x^2+a^2)^{\frac{3}{2}}}$, where total charge $Q = \boldsymbol{\lambda}\times 2\pi a$ ½

At large distance i.e. x>>a

$E \simeq \frac{1}{4\pi\varepsilon_o}\cdot\frac{Q}{x^2}$ ½

This is the Electric field due to a point charge at distance x. ½

(**NOTE**: Award two marks for this question, if a student attempts this question but does not give the complete answer)